

SSB Transmission

Unit objective

At the completion of this unit, you will be able to describe single sideband (SSB) transmission circuits by using the AM/SSB TRANSMITTER circuit block on the ANALOG COMMUNICATIONS circuit board.

Unit fundamentals

SSB Transmission: Fundamentals

AM Transmission described and demonstrated Amplitude Modulation (AM) transmission.

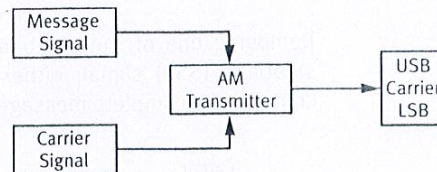


Figure 468

The transmitter converts the message signal frequency to the frequency of the carrier signal and transmits the upper sideband (USB), lower sideband (LSB), and the carrier.

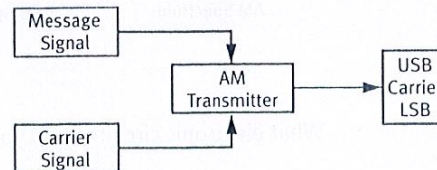


Figure 469

For an AM signal, the message signal is frequency translated to sidebands centered around the carrier frequency.

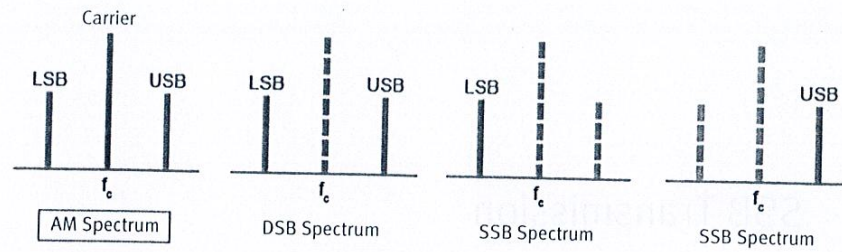


Figure 470

Double sideband (DSB) modulation removes the carrier frequency (f_c) from the AM signal, leaving only the LSB and USB frequencies.

A DSB signal is an amplitude-modulated signal with the carrier frequency removed.

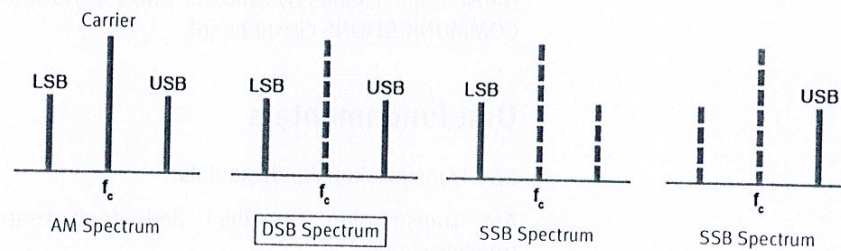


Figure 471

Removing one of the sideband frequencies from a DSB signal produces a single sideband (SSB) signal; either sideband may be removed because each sideband contains the complete message signal intelligence.

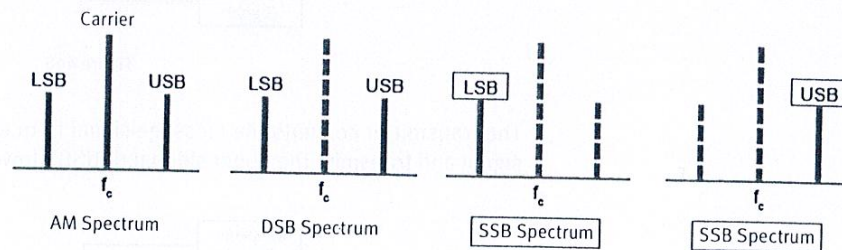


Figure 472

What electronic circuit would you use to produce an SSB from a DSB?

- balanced modulator
- diode detector
- filter

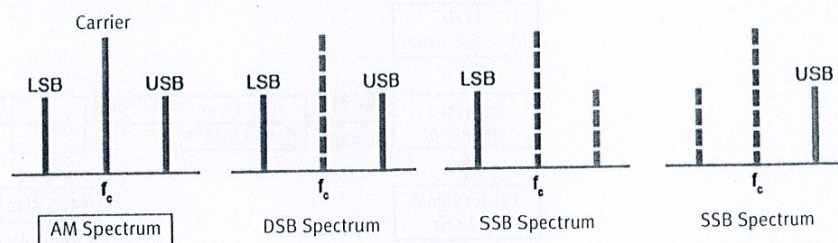


Figure 473

On the circuit board, a balanced modulator translates the 3 kHz message signal frequency and suppresses the 452 kHz carrier signal in the SSB transmitter.

The 449 kHz LSB and the 455 kHz USB are output from the balanced modulator.

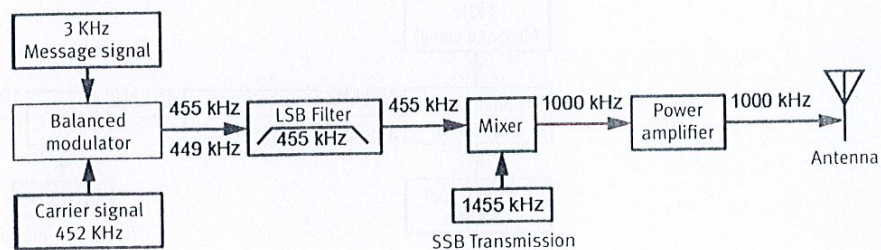


Figure 474

A ceramic filter attenuates the 449 kHz LSB, and the 455 kHz USB is output as the SSB signal to the balanced mixer.

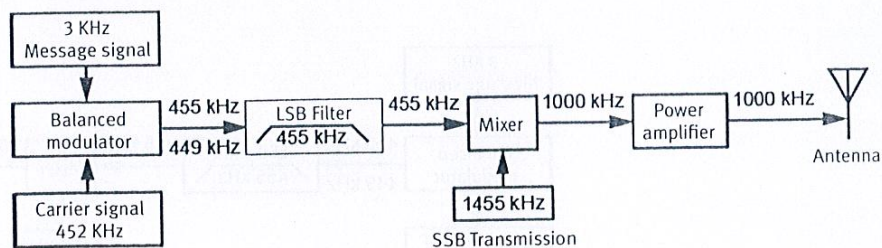


Figure 475

The mixer, which is a balanced modulator, increases the frequency of the 455 kHz SSB signal to the transmission frequency of 1000 kHz.

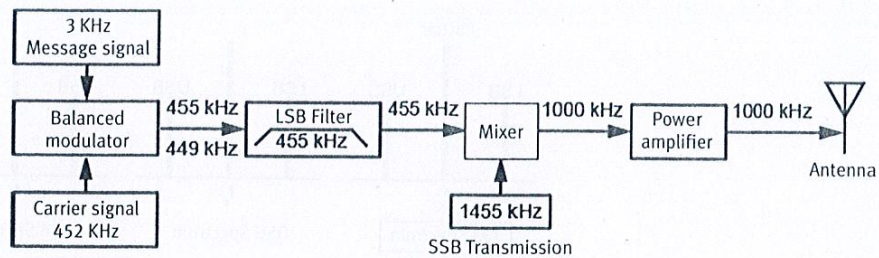


Figure 476

A tuned LC network at the output of the mixer attenuates the 455 kHz input and 1910 kHz sum frequencies; the 1000 kHz SSB signal is output from the mixer to a power amplifier.

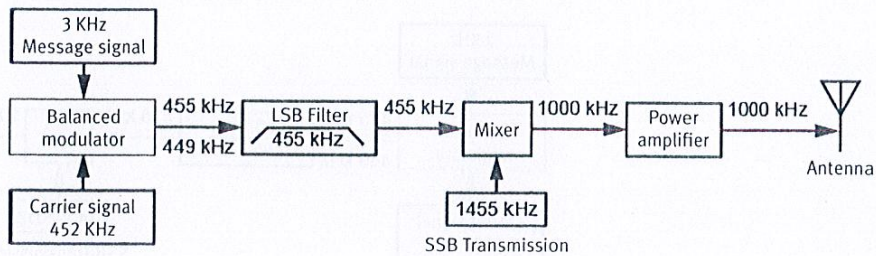


Figure 477

A high gain power amplifier increases the SSB power to the antenna.

On your circuit board, the power amplifier and its antenna matching network tuned for 1000 kHz are the same circuits used for AM transmission.

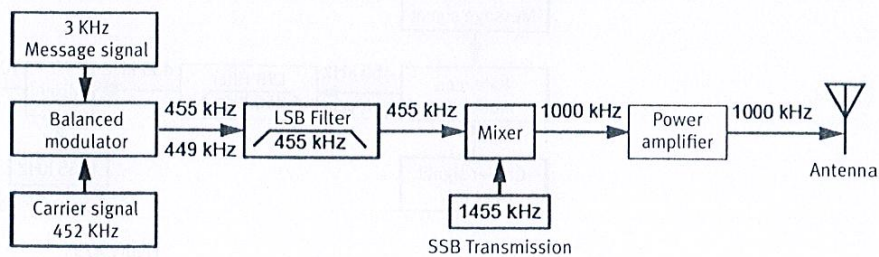


Figure 478

What are the advantages of SSB transmission over AM transmission?

- a. Fewer electronic components are required.
- b. Lower power and a smaller bandwidth are required.